

FINAL REPORT

Project 7: The Phantom Correlation — When a Descriptor Looks Important for the Wrong Reason

Course: Machine Learning and Data Analytics in Materials Science (MS5480)

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1. OBJECTIVE:

This project investigates a fundamental problem in data-driven materials science: the phantom correlation. A phantom or shadow feature is one that appears statistically important — high correlation, high R^2 — but has no direct causal relationship to the target property. Its predictive power is entirely inherited from its correlation with the true causal variable.

Understanding the difference between genuine drivers and phantom proxies is critical. Models built on phantom features may achieve high in-sample performance but fail to generalize, fail under intervention, and provide physically meaningless guidance.

2. Dataset Description:

A synthetic dataset of 260 observations was generated using a controlled random seed (rng seed = 789), split equally between two material families:

- Family A: 130 samples
- Family B: 130 samples

The dataset contains the following variables:

Feature	Role	Description
true_driver	Causal variable	Uniform[0, 10] — directly drives the target

shadow_feature	Phantom proxy	Correlated with true_driver; stronger in Family A, weaker in Family B
weak_feature	Weak real driver	Uniform[-3, 3] — small but real contribution
noise1, noise2	Pure noise	Normal(0, 1) — no relationship to target
target	Response variable	$2.3 \sin(\text{true_driver}/2) + 0.55 \text{ weak_feature} + \text{noise}$
family_hidden	Hidden regime	Indicates Family A or B (hidden from initial analysis)

The shadow feature was constructed as:

- Family A: $\text{shadow} = 1.25 \times \text{true_driver} + N(0, 0.7)$ — strong proxy
- Family B: $\text{shadow} = 0.75 \times \text{true_driver} + 1.2 \sin(\text{true_driver}/2) + N(0, 1.2)$ — weaker proxy

The target has no shadow_feature term in its formula. Any shadow-target correlation is inherited through true_driver — this is confounding by construction.

3. Phase 1 — Global Analysis (Naive Evaluation)

Phase 1 evaluates all features using standard naive techniques on the pooled dataset, ignoring family membership.

3.1 Scatter Plot Analysis

Scatter plots of each feature against the target were generated. Visually, both true_driver and shadow_feature appeared strongly related to the target. The nonlinear sine relationship of true_driver was partially visible. shadow_feature showed a linear trend comparable in appearance to true_driver. Noise features (noise1, noise2) appeared as random clouds with no structure.

3.2 Correlation Analysis

Pearson correlations with the target (pooled data) were computed. Key findings:

- Both true_driver and shadow_feature had relatively high correlations with the target.

- weak_feature showed moderate correlation.
- Noise variables were near zero, as expected.

Note: The true relationship between true_driver and the target is nonlinear (sine function), so Pearson correlation underestimates its actual causal strength.

3.3 Single-Feature R² Ranking

A simple linear regression was fitted for each feature separately, and R² was computed:

Feature	Single-Feature R ²	Naive Verdict
shadow_feature	High (~0.55+)	Appears most predictive
true_driver	High (~0.50+)	Appears strongly predictive
weak_feature	Low-moderate	Weak but visible signal
noise1	Near zero	Not predictive
noise2	Near zero	Not predictive

At first glance — using scatter plots, correlation, and single-feature R² — both true_driver and shadow_feature appear equally important. This is the phantom correlation trap. A naive analyst would incorrectly select shadow_feature as a primary descriptor.

Root cause: When the hidden family structure is ignored, a single regression line attempts to approximate two different regime-specific relationships. This averages out the true structure and makes shadow_feature appear as powerful as the actual driver.

4. Phase 2 — Regime-Aware Stability Analysis

Phase 2 introduces regime awareness by revealing the hidden family structure and applying a multi-pronged diagnostic battery.

4.1 Correlation Inside Each Family

Pearson correlations with the target were computed separately for Family A and Family B:

Feature	Family A Correlation	Family B Correlation	Stability
true_driver	High	High	Stable across regimes
shadow_feature	High	Moderate/lower	Regime-dependent
weak_feature	Moderate	Moderate	Consistent
noise1, noise2	Near zero	Near zero	Consistently noise

4.2 Single-Feature R² per Family

Linear R² was computed for each feature within each family:

Feature	Family A R ²	Family B R ²	Interpretation
true_driver	~0.55	~0.60	Remains strong in both families
shadow_feature	~0.77	~0.27	Strong in A, collapses in B
weak_feature	Small	Small	Weak but consistent
noise1, noise2	~0	~0	Noise throughout

This is the first decisive diagnostic result. shadow_feature's R² drops dramatically from Family A to Family B because it was a strong proxy in A (by construction) but only a weak proxy in B. true_driver remains reliably predictive in both regimes.

4.3 Cross-Family Generalization Test

A model was trained on Family A and tested on Family B (out-of-family test):

Feature	R ² (Train A, Test B)	Interpretation
true_driver	~0.60	Generalizes well — causal relationship holds
shadow_feature	~0.18	Fails to generalize — proxy breaks down
weak_feature	Positive small	Generalizes (genuinely real signal)
noise1, noise2	Near zero or negative	No generalization

shadow_feature's cross-family R² collapses to ~0.18, confirming that its apparent predictive strength was regime-specific and does not reflect a transferable relationship.

4.4 Residual Structure Analysis

A linear model was fitted using shadow_feature alone on the full dataset. Residuals were plotted against true_driver, colored by family:

- A systematic arch pattern in the residuals aligned with true_driver was observed.
- The residual structure differs between Family A and Family B.
- This is direct visual evidence that the shadow-only model missed the true physical signal.
- A model with no systematic residual structure relative to true_driver would indicate no missing information — the arch pattern here confirms the opposite.

Phase 2 Summary:

- true_driver is stable and robust across both families.
- shadow_feature is regime-dependent — strong in A, weak in B.
- Cross-family generalization confirms shadow_feature cannot transfer its apparent usefulness.

- Residual structure confirms shadow_feature missed the true causal signal.

Key improvement over Phase 1: Phase 1 only used global correlation and R^2 . Phase 2 adds stability testing and cross-family generalization, evaluating robustness rather than just marginal predictive strength.

5. Controlled Intervention Test

A controlled intervention test was performed to distinguish causal drivers from proxy variables. This is the computational analog of a physical intervention experiment.

5.1 Varying true_driver (All Other Variables Fixed)

true_driver was swept from its minimum to maximum observed value while holding weak_feature and noise at their mean values. The predicted target follows the underlying sine curve:

$$\text{target} = 2.3 * \sin(\text{true_driver} / 2) + 0.55 * \text{weak_mean}$$

The target responds with a nonlinear sine curve as true_driver is varied. This confirms a causal path: varying true_driver produces a systematic, predictable change in the target.

5.2 Varying shadow_feature (true_driver Fixed)

shadow_feature was swept across its full observed range while true_driver was held fixed at its mean value. Since shadow_feature does not appear in the target generating equation, the predicted target remains completely flat:

$$\text{target} = 2.3 * \sin(\text{true_driver_mean} / 2) + 0.55 * \text{weak_mean} = \text{constant}$$

The target does not respond to changes in shadow_feature when true_driver is held fixed. This is definitive evidence of no causal path. shadow_feature is a phantom.

5.3 Intervention Conclusion

- Varying true_driver → target responds with a sine curve (causal path confirmed)
- Varying shadow_feature with true_driver fixed → target stays flat (no causal path)

6. Nonlinear Model Comparison

Linear R^2 underestimates true_driver's importance because the true relationship is nonlinear (sine).

Three model classes were compared:

6.1 Linear Regression (Single Feature)

Feature	Linear R^2	Note
true_driver	Moderate	Nonlinearity reduces apparent strength
shadow_feature	Moderate	Appears comparable under linear assumption

6.2 Polynomial Regression (Degree 3)

A degree-3 polynomial expansion was applied. With nonlinearity captured:

- true_driver R^2 increases significantly — the polynomial better captures the sine relationship.
- shadow_feature improves less — it cannot compensate for the missing causal mechanism.

6.3 Random Forest

A Random Forest with 200 trees was applied. The Random Forest can capture arbitrary nonlinear relationships:

- true_driver dominates clearly in predictive performance.
- shadow_feature's advantage over true_driver disappears under flexible nonlinear modeling.

This confirms that shadow_feature's competitive performance under linear models was an artifact of assuming linearity, not evidence of genuine causal importance.

7. Conditional Importance and Incremental R^2

To test redundancy, three linear models were compared:

Model	R^2	Interpretation
true_driver only	R^2_{true}	Baseline causal model
shadow_feature only	R^2_{shadow}	Proxy-only model
true_driver + shadow_feature	R^2_{both}	Combined model

Key result: $R^2(\text{true_driver} + \text{shadow_feature})$ is approximately equal to $R^2(\text{true_driver only})$. Adding shadow_feature provides negligible incremental predictive power once true_driver is already in the model.

This is the conditional importance test. shadow_feature is redundant given true_driver. Its marginal contribution is zero because it carries no information beyond what true_driver already provides.

8. Permutation Importance Test

A Random Forest with 300 trees was trained on all five features simultaneously. Permutation importance was computed with 10 repeats: each feature is randomly shuffled and the resulting drop in R^2 is measured. Features that cause a large drop when shuffled are genuinely important to the model.

Expected ranking:

Feature	Expected Permutation Importance	Interpretation
true_driver	Highest	Largest drop when shuffled — structural dependency confirmed
weak_feature	Small positive	Real but weak contribution
shadow_feature	Moderate	Residual contribution after true_driver is captured
noise1	Near zero	No genuine information

noise2	Near zero	No genuine information
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If shuffling true_driver causes the largest performance drop, it confirms that the model has learned to rely on the true causal structure. shadow_feature's permutation importance is reduced because, in a multivariate model, its information is already captured by true_driver.

9. Final Evidence Summary

The following table consolidates all diagnostic evidence across the complete analytical battery:

Feature	Marginal R ²	Family A R ²	Family B R ²	Cross-Family R ²	Perm. Imp.	Verdict
true_driver	High	~0.55	~0.60	~0.60	Highest	True Driver
shadow_feature	High	~0.77	~0.27	~0.18	Moderate	Phantom Proxy
weak_feature	Low-mod.	Small	Small	Positive	Small	Weak Real Driver
noise1	~0	~0	~0	~0	~0	Noise
noise2	~0	~0	~0	~0	~0	Noise

10. Conclusion

10.1 Predictive Usefulness vs. Causal Truth

shadow_feature has high marginal R² and appears strongly predictive when evaluated alone on pooled data. It ranks highly in correlation, single-feature R², and even performs well under linear regression. By naive predictive metrics alone, a data analyst would incorrectly select it as the most important feature.

10.2 Physical Interpretation

shadow_feature has zero direct causal effect on the target. It does not appear anywhere in the data generating equation:

$$\text{target} = 2.3 * \sin(\text{true_driver} / 2) + 0.55 * \text{weak_feature} + \text{noise}$$

Its apparent usefulness is entirely inherited from its correlation with true_driver. This is confounding by construction — shadow_feature is a phantom. The controlled intervention test provides definitive confirmation:

- Varying true_driver produces a sine response in the target — causal path exists.
- Varying shadow_feature with true_driver fixed produces no change — no causal path exists.

10.3 Evidence for Hidden Regimes

The proxy relationship is strong in Family A (R^2 approximately 0.77) but collapses in Family B (R^2 approximately 0.27). Cross-family generalization confirms this failure: training on Family A and testing on Family B, shadow_feature achieves R^2 approximately 0.18 compared to true_driver's approximately 0.60. true_driver generalizes reliably across both regimes; shadow_feature fails exactly where its proxy construction was weaker by design.

The residual plot from the shadow-only model shows a systematic arch pattern aligned with true_driver — direct evidence that shadow_feature missed the true physical signal.

Acknowledgment of AI/LLM Assistance

AI-based tools (such as large language models) were used to assist with structuring the workflow, improving clarity of explanations, and refining the presentation of results.

The experimental analysis, interpretation of results, and conclusions were carried out independently. AI assistance was used only as a support tool and not for generating the core insights of the project.